

A Larger Three-Dimensional Gap for the Lattice Schaffer Constants

Agent packet for arXiv:2505.16775

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Status

This is a candidate partial result for Rincon-Villamizar and Oikhberg, *The lattice Schaffer constant*, arXiv:2505.16775. It improves the displayed lower bounds in the source paper's question about the maximal possible values of $\beta(X)/\lambda^+(X)$ and $\beta(X) - \lambda^+(X)$. It does not determine the maxima.

Source Question

The source paper defines, for a Banach lattice X ,

$$\lambda^+(X) = \inf\{\|x + y\| : x, y \in S_X^+\}$$

and

$$\beta(X) = \inf\{\|x \vee y\| : x, y \in S_X^+, x \wedge y = 0\}.$$

Remark 3.7 asks for the maximal possible values of the ratio $\beta(X)/\lambda^+(X)$ and the difference $\beta(X) - \lambda^+(X)$. The relevant source-page crop is included below.

Remark 3.7. Clearly $1 \leq \lambda^+(X) \leq \beta(X) \leq 2$ for any Banach lattice X , and Example 3.5 shows that the middle inequality can be strict. We do not know the maximal values of $\beta(X)/\lambda^+(X)$ or $\beta(X) - \lambda^+(X)$.

Now we show examples of $\lambda^+(X)$ for some particular Banach lattices X .

Idea of the Proof

The construction keeps the norm very small on two positive unit vectors that share the third coordinate, forcing $\lambda^+ \leq 4/3$. At the same time, disjoint positive vectors cannot share that third coordinate. The disjoint case then reduces to one variable and forces a cost at least $3/2$.

The lower bound for λ^+ is a short averaging argument. For each positive unit vector (x, y, z) , the auxiliary quantity

$$H(x, y, z) = x + y + \frac{2z}{3}$$

is at least 1. For a sum of two positive unit vectors, the three defining functionals of the norm add up to $2H$, so one of them must be at least $4/3$.

Result

Theorem 1. *Let $X = \mathbb{R}^3$ with coordinatewise order and lattice norm*

$$\|(a, b, c)\| = \max \left\{ |a| + \frac{2}{3}|c|, |b| + \frac{2}{3}|c|, |a| + |b| \right\}.$$

Then

$$\lambda^+(X) = \frac{4}{3}, \quad \beta(X) = \frac{3}{2}.$$

Consequently

$$\frac{\beta(X)}{\lambda^+(X)} = \frac{9}{8}, \quad \beta(X) - \lambda^+(X) = \frac{1}{6}.$$

Proof. The displayed formula is the maximum of positive linear functionals applied to $(|a|, |b|, |c|)$, so it is an absolute monotone norm and therefore a lattice norm for the coordinatewise order.

For $u = (x, y, z) \in X^+$, write

$$A(u) = x + \frac{2z}{3}, \quad B(u) = y + \frac{2z}{3}, \quad C(u) = x + y.$$

Then $\|u\| = \max\{A(u), B(u), C(u)\}$. Also set

$$H(u) = x + y + \frac{2z}{3}.$$

If $u \in S_X^+$, at least one of $A(u), B(u), C(u)$ is equal to 1, and each of $A(u), B(u), C(u)$ is bounded above by $H(u)$. Hence

$$H(u) \geq 1 \quad (u \in S_X^+).$$

Let $u, v \in S_X^+$, and put $w = u + v = (X_1, X_2, X_3)$. Then

$$H(w) = H(u) + H(v) \geq 2.$$

Moreover

$$A(w) + B(w) + C(w) = \left(X_1 + \frac{2X_3}{3}\right) + \left(X_2 + \frac{2X_3}{3}\right) + (X_1 + X_2) = 2H(w) \geq 4.$$

Thus at least one of $A(w), B(w), C(w)$ is at least $4/3$, and therefore

$$\|u + v\| \geq \frac{4}{3}.$$

This proves $\lambda^+(X) \geq 4/3$. Conversely,

$$u_0 = \left(\frac{2}{3}, 0, \frac{1}{2}\right), \quad v_0 = \left(0, \frac{2}{3}, \frac{1}{2}\right)$$

are positive unit vectors, and

$$\|u_0 + v_0\| = \left\| \left(\frac{2}{3}, \frac{2}{3}, 1\right) \right\| = \frac{4}{3}.$$

Therefore $\lambda^+(X) = 4/3$.

It remains to compute $\beta(X)$. The upper bound is witnessed by the disjoint positive unit vectors

$$p_0 = (1, 0, 0), \quad q_0 = \left(0, \frac{1}{2}, \frac{3}{4}\right),$$

for which

$$\|p_0 + q_0\| = \left\| \left(1, \frac{1}{2}, \frac{3}{4}\right) \right\| = \frac{3}{2}.$$

Hence $\beta(X) \leq 3/2$.

For the reverse inequality, let $p, q \in S_X^+$ be disjoint. Since there are only three coordinates, after interchanging p and q , and after interchanging the first two coordinates if needed, the disjoint supports fall into the following cases.

First suppose neither vector uses the third coordinate. Then the two vectors are supported on disjoint subsets of the first two coordinates. Positive unit vectors on the coordinate axes are $(1, 0, 0)$ and $(0, 1, 0)$, so $\|p + q\| = 2$.

Next suppose one vector is supported only on the third coordinate. It must be $(0, 0, 3/2)$. The other vector has the form $(a, b, 0)$ with $a, b \geq 0$ and $a + b = 1$. Hence

$$\|(a, b, 3/2)\| \geq \max\{a + 1, b + 1\} = 1 + \max\{a, b\} \geq \frac{3}{2}.$$

Finally suppose one vector uses the third coordinate and exactly one of the first two coordinates, while the other vector uses the remaining first coordinate. By symmetry, write

$$p = (a, 0, c), \quad q = (0, 1, 0),$$

with $a, c \geq 0$. Since $\|p\| = 1$, we have

$$a + \frac{2c}{3} = 1.$$

Therefore

$$\|p + q\| = \|(a, 1, c)\| = \max \left\{ 1, 1 + \frac{2c}{3}, 1 + a \right\}.$$

Writing $2c/3 = 1 - a$, this becomes

$$\max\{1, 2 - a, 1 + a\} \geq \frac{3}{2}.$$

These cases exhaust all disjoint positive unit pairs in $\mathbb{R}^3$. Therefore $\beta(X) \geq 3/2$, and the previously shown upper bound gives $\beta(X) = 3/2$. $\square$

Comparison With the Source Example

The source paper's Example 3.5 gives a three-dimensional lattice with $\beta = 15/11$ and $\lambda^+ \leq 4/3$, followed by Remark 3.7 asking for the maximal possible ratio and difference. The example above gives the exact larger lower bounds

$$\sup_X \frac{\beta(X)}{\lambda^+(X)} \geq \frac{9}{8}, \quad \sup_X (\beta(X) - \lambda^+(X)) \geq \frac{1}{6}.$$

The packet does not claim these lower bounds are optimal.

Verification Notes

The script `code/verify_constants_and_crop.py` enumerates active faces of the positive unit ball for the polyhedral norm and independently recovers $\lambda^+ = 4/3$ and $\beta = 3/2$. This computation is only a check; the proof above is the certificate.

Cheap local index searches were run for 2505.16775, lattice Schaffer keywords, and β/λ -type terms. No prior packet for this target was found. No broad external web search was performed for this partial result.

References

References

- [1] M. A. Rincon-Villamizar and T. Oikhberg, *The lattice Schaffer constant*, arXiv:2505.16775.